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Microscope unit

Abstract

A microscope unit comprises: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source, the illumination lens barrel having: a first lens barrel configured capable of being fitted with the light source; and an intermediate lens barrel connecting the main lens barrel and the first lens barrel, and a field stop of light irradiated from the light source being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel is.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of Japanese Patent Application No. 2021-088019, filed on May 25, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

Field

(2) The present invention relates to a microscope unit capable of being fitted to a target device.

Description of the Related Art

(3) Up to now, there has been known a microscope unit comprising: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source. Such a microscope unit is configured so that, for example, a light flux (a flux of diverging light rays) that has been emitted from the light source is collected by a collector lens, passes through a relay lens system and field stop to be reflected by a mirror, and passes through a relay lens system, aperture stop, and condenser lens to illuminate an observation surface.

(4) Such a microscope unit is sometimes capable of being fitted with a variety of light sources. However, it has sometimes been the case that, depending on a kind of light source fitted to the microscope unit, illumination at the observation surface ends up getting ununiform. Accordingly, for example, in a microscope illumination device described in Japanese Patent Application Publication No. 2002-6225, a fly-eye lens is disposed in the illumination optical system, whereby uniformity of illumination at the observation surface is achieved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic explanatory diagram showing overall configuration of a microscope unit according to a first embodiment.

(2) FIG. 2 is a schematic explanatory diagram showing overall configuration of the microscope unit.

(3) FIG. 3 is a schematic side view showing configuration of a mirror frame member of the microscope unit.

(4) FIG. 4 is a schematic perspective view showing configuration of the mirror frame member of the microscope unit.

(5) FIGS. 5a and 5b are diagrams for explaining a relationship between attitude of the illumination lens barrel with respect to the main lens barrel of the microscope unit and illumination range.

(6) FIGS. 6a, 6b and 6c are diagrams for explaining illumination range in the case of fitting position adjustment having been completed and the case of fitting position adjustment not yet having been performed.

(7) FIGS. 7a and 7b are schematic explanatory diagrams showing configuration of a microscope unit according to a second embodiment.

(8) FIGS. 8a and 8b are explanatory diagrams showing outward appearance of the microscope unit.

(9) FIGS. 9a, 9b and 9c are explanatory diagrams showing outward appearance of the microscope unit.

(10) FIGS. 10a, 10b and 10c are explanatory diagrams showing outward appearance of the microscope unit.

DETAILED DESCRIPTION

(11) In the above-described microscope illumination device, the fly-eye lens needs to be provided in the illumination optical system. Moreover, a relay lens needs to be provided between the fly-eye lens and the field stop in order to relay light from the fly-eye lens. There is a risk that this will end up leading to an increase in size of the illumination lens barrel and, consequently, end up leading to an increase in size of the microscope unit.

(12) The present invention, which was made in view of such a problem, has an object of providing

a microscope unit that enables downsizing of an illumination lens barrel overall to be achieved, and, consequently, enables downsizing of the microscope unit to be achieved.

(13) A microscope unit according to one embodiment of the present invention comprises: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source, the illumination lens barrel having: a first lens barrel configured capable of being fitted with the light source; and an intermediate lens barrel connecting the main lens barrel and the first lens barrel, and a field stop of light irradiated from the light source being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel is.

(14) In this kind of microscope unit, the field stop of the illumination lens barrel is disposed more inwardly than the outer peripheral surface of the intermediate lens barrel is. Hence, it is possible for space inside the first lens barrel to be utilized more effectively compared to in such a case as when, for example, the field stop is disposed more outwardly than the outer peripheral surface of the intermediate lens barrel is. This makes it possible for total length of the illumination lens barrel to be configured short, whereby downsizing of the illumination lens barrel overall is achieved, and, consequently, downsizing of the microscope unit is achieved.

(15) The above-described field stop may be disposed more inwardly than an inner peripheral surface of the intermediate lens barrel is.

(16) The above-described intermediate lens barrel may be connected to the main lens barrel so as to freely revolve around its own optical axis with respect to the main lens barrel.

(17) The above-described microscope unit may have: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member. Moreover, in such a case, the field stop may be disposed more inwardly than an outer peripheral surface of the mirror frame member is.

(18) Moreover, the illumination optical system may have: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays to the beam splitter light that has been reflected by the reflecting mirror. In such a case, there may be adopted a configuration of a kind that the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.

(19) Moreover, the field stop may be part of the first lens barrel, and may be housed in the intermediate lens barrel.

(20) Such a configuration makes it possible to provide a microscope unit that enables downsizing of an illumination lens barrel overall to be achieved, and, consequently, enables downsizing of the microscope unit to be achieved.

(21) Microscope units according to several embodiments of the present invention will be described in detail below with reference to the accompanying drawings. However, the embodiments below are not to limit the inventions according to each of the claims, moreover, not all of the combinations of features described in the embodiments are necessarily essential to the means for solving the problem of the invention. Note that in the embodiments below, scales, dimensions, and

so on of each of configuring elements are sometimes shown expanded, and some of the configuring elements are sometimes omitted.

(22) Moreover, in the description below, an “X axis direction” means a left-right direction in the case of having correctly faced the paper surface (front surface) of a microscope unit shown in FIG. 1, a “Y axis direction” means a depth direction in same case, and a “Z axis direction” means an up-down direction, that is, an optical axis direction of a main lens barrel in same case. Note that basic structure of a microscope unit is already known, so, apart from where necessary, overall configuration will be described here in outline only.

First Embodiment

(23) [Configuration]

(24) FIGS. 1 and 2 are schematic explanatory diagrams showing overall configuration of a microscope unit **100** according to a first embodiment. In FIGS. 1 and 2, part of the microscope unit **100** is shown as a cross-sectional view, and part of the microscope unit **100** is shown as a transparency view. FIG. 2 shows a situation where attitude with respect to a main lens barrel **10** of an illumination lens barrel **20** differs by 180° compared to in FIG. 1. Hereafter, description will be made assuming a state shown in FIG. 1 to be a basic attitude of the microscope unit **100**.

(25) As shown in FIGS. 1 and 2, the microscope unit **100**, which configures part of an unillustrated microscope, comprises: the main lens barrel **10**; and the illumination lens barrel **20** which is connected to the main lens barrel **10**.

(26) The main lens barrel **10**, which configures a so-called imaging optical system of the microscope, is for example configured so that its end portion on an upper side in the Z axis direction exemplified in FIG. 1 (hereafter, simply called “upper side”) is capable of being fitted with a camera **31** incorporating an imaging sensor **30**, and its end portion on a lower side in the Z axis direction exemplified in FIG. 1 (hereafter, simply called “lower side”) is capable of being fitted with an objective lens **32**. Note that the imaging sensor **30** includes various kinds of image sensors (CMOS, CCD), and so on.

(27) The main lens barrel **10** has: a first sleeve portion **11** of cylindrical shape disposed on the upper side (a camera **31** side); and a second sleeve portion **12** of cylindrical shape disposed on a lower side of this first sleeve portion **11**. The first sleeve portion **11** is formed so that its minimum outer diameter will be slightly smaller than a minimum outer diameter of the second sleeve portion **12**.

(28) Moreover, the main lens barrel **10** has: a case of main body portion **13** of rectangular-shaped outward appearance disposed on a lower side of the second sleeve portion **12**; and a third sleeve portion **14** of cylindrical shape disposed on a lower side of this case of main body portion **13** and being connected with the objective lens **32**. Note that the case of main body portion **13** is configured by: a first casing **15**; and a second casing **16** which is disposed on a lower side of this first casing **15**.

(29) On insides of the second sleeve portion **12** and the first casing **15** of the case of main body portion **13** in the main lens barrel **10**, there is housed an imaging lens (a tube lens) **17**. Moreover, on an inside of the second casing **16** of the case of main body portion **13** in the main lens barrel **10**, a beam splitter (hereafter, written as “B/S”) **18** that will be mentioned later is housed so as to incline at a certain angle to an optical axis ML of the main lens barrel **10**. These first sleeve portion **11**, second sleeve portion **12**, case of main body portion **13**, and third sleeve portion **14** configuring the main lens barrel **10** are coaxially disposed linearly in the Z axis direction, along the optical axis ML of the main lens barrel **10**.

(30) The illumination lens barrel **20**, which configures a so-called illumination optical system of the microscope, has: a first lens barrel **21** capable of having a variety of light sources LS fitted in a freely attachable/detachable manner to its end portion on the upper side in the Z axis direction in the basic attitude; and an intermediate lens barrel **22** that connects this first lens barrel **21** to the main lens barrel **10**. Note that the first lens barrel **21** has its optical axis L1 formed in the Z axis

direction, and the intermediate lens barrel **22** has its optical axis L2 formed in the X axis direction.

(31) The first lens barrel **21** has: a fourth sleeve portion **23** of cylindrical shape disposed on the upper side in the basic attitude; and a fifth sleeve portion **24** of cylindrical shape disposed on a lower side of this fourth sleeve portion **23**. Note that the fourth sleeve portion **23** is formed so that its minimum outer diameter will be smaller than a minimum outer diameter of the fifth sleeve portion **24**.

(32) Moreover, the first lens barrel **21** has: a coupler case of portion **26** of stepped annularly shaped outward appearance disposed on a lower side of the fifth sleeve portion **24**; and a sixth sleeve portion **27** of stepped cylindrically shaped outward appearance disposed on a lower side of the coupler case of portion **26**. The coupler case of portion **26** couples the fifth sleeve portion **24** and the sixth sleeve portion **27**. Note that part of the sixth sleeve portion **27** (a portion thereof disposed within the intermediate lens barrel **22**) configures a truncated conical portion **27a** which is formed in a downwardly tapering truncated conical shape. Although in the example illustrated, outer diameters of the fifth sleeve portion **24** and the sixth sleeve portion **27** are substantially the same and a minimum outer diameter of the coupler case of portion **26** is shown larger than these outer diameters, outer diameters of the fifth sleeve portion **24**, the coupler case of portion **26**, and the sixth sleeve portion **27** may be configured to be substantially the same.

(33) On an inside of the fourth sleeve portion **23** in the first lens barrel **21**, there is housed a light-emitting end LE of the fitted light source LS. Moreover, the fifth sleeve portion **24** in the first lens barrel **21** houses on its inside on the upper side a collector lens (a light-collecting lens) **33**. Furthermore, on an inside of a connecting point of the fifth sleeve portion **24** and the coupler case of portion **26**, there is housed a fly-eye lens **34**.

(34) The sixth sleeve portion **27** in the first lens barrel **21** houses on its inside on the upper side a first relay lens group **35**. The first relay lens group **35** has a plurality of lenses that are not joined to each other. For example, in the example illustrated, the first relay lens group **35** comprises: a first lens **35a** consisting of a plano-convex lens having its convex surface directed to the lower side in the Z axis direction; and a second lens **35b** consisting of a plano-convex lens having its convex surface directed to the upper side in the Z axis direction. These first lens **35a** and second lens **35b** are disposed so that fellow convex surfaces face each other. The first lens **35a** and the second lens **35b** may be disposed separated from each other, or may have parts of their convex surfaces in contact with each other.

(35) Note that an inner peripheral surface **27b** more to the lower side than a housing place of the first relay lens group **35** of the sixth sleeve portion **27** in the first lens barrel **21** is formed in a tapered shape whose diameter gradually decreases in a downward direction. Moreover, part of an outer peripheral portion of the sixth sleeve portion **27** has formed therein a flange portion **27c** of annular shape projecting in an outer peripheral direction.

(36) Further to the lower side than a lower end surface **27d** of the flange portion **27c** of the sixth sleeve portion **27** in the first lens barrel **21**, an outer peripheral portion **27e** whose diameter is smaller than a minimum outer diameter of the sixth sleeve portion **27** is passed, and a stepped portion formed still further below is traversed, whereat there is disposed an outer peripheral portion **27f** of the truncated conical portion **27a**, that is formed in a tapered shape whose diameter gradually decreases in a downward direction.

(37) Note that the fourth sleeve portion **23**, fifth sleeve portion **24**, coupler case of portion **26**, and sixth sleeve portion **27** configuring the first lens barrel **21** are linearly coaxially disposed along the optical axis L1 of the first lens barrel **21**. Moreover, a lower end portion of the truncated conical portion **27a** of the sixth sleeve portion **27** configures a field stop **36**. That is, the field stop **36** in the illumination lens barrel **20** according to the first embodiment is configured by part of the first lens barrel **21** (a lower end portion of the truncated conical portion **27a**).

(38) The intermediate lens barrel **22** connects the first lens barrel **21** and the main lens barrel **10** and has an intermediate case of portion **28** of rectangular outward appearance that, in the basic

attitude, has formed therein on a side of its end portion separated in the X axis direction from the main lens barrel **10** an inclined wall **28a** inclining obliquely downwardly from the upper side toward the main lens barrel **10**.

(39) Note that an end portion on a side of the main lens barrel **10** in the X axis direction of the intermediate case of portion **28** is formed as a mating cylindrical portion **28b** of cylindrical shape, and is mated with a mating portion **16a** which is hole-like, formed in the second casing **16** of the case of main body portion **13**. As a result, the intermediate lens barrel **22** is connected to the main lens barrel **10** so as to be capable of revolving around the optical axis L2 orthogonal to the optical axis L1 of the first lens barrel **21**, with respect to the main lens barrel **10** (refer to FIG. 2). Note that a second relay lens **38** is housed on an inside of the mating cylindrical portion **28b**.

(40) Moreover, an upper wall portion **28c** of the intermediate case of portion **28** in the basic attitude has formed therein a mating hole **28d** with which the sixth sleeve portion **27** of the first lens barrel **21** is mated. That is, the sixth sleeve portion **27** is mated with the intermediate case of portion **28** in such a state that the lower end surface **27d** of the flange portion **27c** abuts on an outer wall surface of the upper wall portion **28c** and the outer peripheral portion **27e** catches on an inner peripheral surface on an opening end side of the mating hole **28d**.

(41) Note that the sixth sleeve portion **27** is fixed to the intermediate case of portion **28** by the flange portion **27c** being fitted to the upper wall portion **28c** by fitting bolts **19**. As a result, the first lens barrel **21** is fixed after having been positioned with respect to the intermediate lens barrel **22**. This configuration results in the first lens barrel **21** being fixed in this way to the intermediate lens barrel **22**, so leads to the above-mentioned field stop **36** of the first lens barrel **21** being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel **22** (the outer wall surface of the upper wall portion **28c** of the intermediate case of portion **28**) is.

(42) Moreover, the intermediate lens barrel **22** has housed therein: a reflecting mirror **37**; and a mirror frame member **40** that holds this reflecting mirror **37**. Moreover, the intermediate lens barrel **22** is provided with a position adjusting mechanism (not illustrated) for adjusting a position of fitting to the intermediate lens barrel **22** (a position of fitting to the intermediate casing portion **28**) of the mirror frame member **40**.

(43) FIG. 3 is a schematic side view showing configuration of the mirror frame member **40** of the microscope unit **100**. Moreover, FIG. 4 is a schematic perspective view showing configuration of the mirror frame member **40** of the microscope unit **100**.

(44) As shown in FIGS. 3 and 4, the mirror frame member **40** is formed by a frame body **49** that, so as to allow the mirror frame member **40** to be housed inside the intermediate case of portion **28**, has a cylindrical shape of a kind that one end portion side thereof has been obliquely cut. The frame body **49** of the mirror frame member **40** holds the reflecting mirror **37**. This reflecting mirror **37** is formed with an elliptical or polygonal (octagonal, or the like) outer shape, for example, so that its reflecting portion will have an elliptical shape matching a shape of an elliptical opening portion **41** of the frame body **49**. Moreover, a portion excluding the above-described reflecting portion, of this reflecting mirror **37** is adhered to the frame body **49** of the mirror frame member **40** via the likes of an unillustrated adhesive, for example.

(45) Moreover, the frame body **49** of the mirror frame member **40** has: a main lens barrel side opening portion **42** formed at an end portion on an opposite side in the X axis direction to the elliptical opening portion **41**, of the frame body **49**; and a first lens barrel side opening portion **43** formed at a position separated in the Z axis direction from the elliptical opening portion **41**, looking from the elliptical opening portion **41**. The first lens barrel side opening portion **43** has an opening diameter larger than an outer diameter of the field stop **36** of the truncated conical portion **27a** in the sixth sleeve portion **27** of the first lens barrel **21** and larger than an outer diameter of the outer peripheral portion **27f** in a certain range on the upper side in the Z axis direction from the field stop **36**. As a result, the field stop **36** can be disposed brought as close as is possible to the reflecting mirror **37**, without the frame body **49** of the mirror frame member **40** interfering with the truncated

conical portion **27a** of the sixth sleeve portion **27**.

(46) Note that after the mirror frame member **40** thus configured has been housed inside the intermediate case of portion **28**, the mirror frame member **40** has its fitting position adjusted with respect to the intermediate case of portion **28** by the unillustrated position adjusting mechanism. Although illustration of the position adjusting mechanism is omitted, it is configured by a plurality of adjustment screws that have been screwed into a pair of seating countersunk holes drilled in the Y axis direction in the intermediate case of portion **28**, for example. After adjustment of the fitting position, the mirror frame member **40** is fixed to the intermediate case of portion **28** by screw fixing, for example.

(47) Note that since the thus formed intermediate lens barrel **22** and the separately assembled first lens barrel **21** can be integrated by a simple structure only of the flange portion **27c** being fixed by screw fixing to the intermediate case of portion **28** by the fitting bolt **19** in a state of the truncated conical portion **27a** having been inserted in the mating hole **28d**, it is possible for the illumination lens barrel **20** to be easily manufactured.

(48) [Optical Systems]

(49) The imaging optical system of the microscope unit **100** according to the present embodiment comprises the objective lens **32** and the imaging lens (tube lens) **17**. Moreover, the illumination optical system of the microscope unit **100** according to the present embodiment comprises the collector lens (light-collecting lens) **33**, the fly-eye lens **34**, the first relay lens group **35**, the field stop **36**, the reflecting mirror **37**, the B/S **18**, and the second relay lens **38**.

(50) The light-emitting end LE of the light source LS, a position of seating of the objective lens **32** (a pupil of the objective lens **32**), and an emitting end surface of the fly-eye lens **34** are in a conjugate relationship. Moreover, these positions are not in a conjugate relationship with the B/S **18** and the reflecting mirror **37**. Moreover, the field stop **36**, a focal position of the objective lens **32**, and a focal position of the imaging lens **17** are in a conjugate relationship.

(51) A light flux (a flux of diverging light rays) from the light-emitting end LE of the light source LS is converted into a collected light flux (a flux of focused light rays) via the collector lens **33**, and passes through the fly-eye lens **34** to be divided into the same number as there are a plurality of lenses configuring the fly-eye lens **34**. As a result, the emitting end surface of the fly-eye lens **34** has formed therein the same number of light source images as there are a plurality of lenses configuring the fly-eye lens **34** (secondary light sources), and light fluxes (fluxes of diverging light rays) emitted for each of the plurality of lenses configuring the fly-eye lens **34** pass through the first relay lens group **35** to be relayed collected overlapping each other on the field stop **36**. Hence, the emitting end surface of the fly-eye lens **34** has formed therein a pseudo surface light source (secondary light source) having substantially uniform light distribution characteristics with little light distribution variation, even when a variety of light sources LS having different light distribution characteristics have been employed.

(52) A light flux (a flux of diverging light rays) that has passed through the field stop **36** along the optical axis L1 and had its range of light stopped down, is reflected so as to become a light flux (a flux of diverging light rays) along the optical axis L2 by the reflecting mirror **37** in the intermediate lens barrel **22**, and, having passed through the second relay lens **38** to undergo light collection, is relayed to be irradiated onto the B/S **18**. Note that the B/S **18** guides at least a part of the light flux (the flux of diverging light rays) that has passed through the second relay lens **38** to be incident on the B/S **18** reflecting it to a side of an unillustrated aperture stop and the objective lens **32**, and allows to be transmitted through the B/S **18** to a side of the imaging lens **17** at least a part of a light flux from an imaging surface IS that has passed through the objective lens **32**, and so on, to be incident on the B/S **18**.

(53) [Rotation of Illumination Lens Barrel **20** and Positioning of Reflecting Mirror **37**]

(54) As mentioned above, in the microscope unit **100** according to the present embodiment, the illumination lens barrel **20** is configured able to revolve with respect to the main lens barrel **10**.

Now, in the case of a position with respect to the B/S **18** of the reflecting mirror **37** not having been suitably adjusted, there is a risk that when the illumination lens barrel **20** is rotated with respect to the main lens barrel **10**, a center position of the field stop **36** will end up deviating from an optical axis of the illumination optical system. Accordingly, the microscope unit **100** according to the present embodiment is configured to allow the position with respect to the B/S **18** of the reflecting mirror **37** to be adjusted. This point will be described below.

(55) FIGS. **5a** and **5b** are diagrams for explaining a relationship between attitude of the illumination lens barrel **20** with respect to the main lens barrel **10** of the microscope unit **100** and illumination range. FIGS. **6a**, **6b** and **6c** are diagrams for explaining illumination range in the case of fitting position adjustment having been completed and the case of fitting position adjustment not yet having been performed. Note that in FIGS. **5a**, **5b**, **6a**, **6b** and **6c**, the outward appearance of the microscope unit **100** differs from that shown in FIGS. **1** and **2**. Moreover, description is made here assuming fitting position and fitting angle of the B/S **18** of the main lens barrel **10** to have already been adjusted.

(56) In the microscope unit **100** in the basic attitude with the optical axis ML of the main lens barrel **10** and the optical axis L1 of the first lens barrel **21** of the illumination lens barrel **20** being in the Z axis direction as shown in FIG. **5a**, a center of an imaging range **51** in the imaging surface IS and center of an illumination range **52** of light from the light source LS that has passed through the illumination lens barrel **20** are fixed at a regular position centered on the optical axis ML of the main lens barrel **10**, that is, at such a position that the center of the imaging range **51** and center of the illumination range **52** will coincide with the optical axis ML, as shown in FIG. **5b**.

(57) In the microscope unit **100** according to the present embodiment, it is possible for the illumination lens barrel **20** to be rotated by 90° around the optical axis L2 with respect to the main lens barrel **10**, for example, from this basic attitude, as shown in FIG. **6a**. As a result, the illumination range **52** rotates by 90° with an image of the imaging range **51** left unchanged.

(58) Now, if, for example, fitting position of the mirror frame member **40** has been adjusted by the above-mentioned position adjusting mechanism, it is possible, as shown in FIG. **6b**, for a state where the center of the imaging range **51** and center of the illumination range **52** coincide with the optical axis ML to be maintained, regardless of there having been a change in attitude indicated by the arrow in the drawing.

(59) On the other hand, in the case of fitting position of the mirror frame member **40** not having been adjusted, the center of the illumination range **52** will end up being in a state of having greatly deviated from the center of the imaging range **51**, as shown in FIG. **6c**, hence illumination distribution by the light source LS in a visual field will end up getting ununiform.

(60) In order to avoid such a situation, in the microscope unit **100** according to the present embodiment, a position of fitting to the intermediate lens barrel **22** of the mirror frame member **40** is adjusted via the position adjusting mechanism. Adjustment of fitting position of the mirror frame member **40** is performed in such a manner that, for example, subsequent to the illumination lens barrel **20** having been assembled, above-mentioned adjustment screws are operated, whereby a disposing position in the X axis direction with respect to the intermediate case of portion **28** of the mirror frame member **40** and a disposing position in a rotation direction centered on the X axis direction are adjusted.

(61) Adjustment is performed so that an amount of eccentricity of the optical axis L2 accompanying a change in attitude of the first lens barrel **21** at a time when the optical axis L1 of light from the light source LS is converted into light of the optical axis L2 by the reflecting mirror **37**, will fall within a certain range. This makes it possible for attitude with respect to the main lens barrel **10** of the illumination lens barrel **20** to be freely changed, without the illumination range **52** due to the illumination optical system deviating.

(62) [Downsizing of Microscope Unit]

(63) Up to now, there has been known a microscope unit comprising: a main lens barrel of an

imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source. Such a microscope unit is configured so that, for example, a light flux (a flux of diverging light rays) that has been emitted from the light source is collected by a collector lens, passes through a relay lens system and field stop to be reflected by a mirror, and passes through a relay lens system, aperture stop, and condenser lens to illuminate an observation surface.

(64) Such a microscope unit is sometimes capable of being fitted with a variety of light sources. However, it has sometimes been the case that, depending on a kind of light source fitted to the microscope unit, illumination at the observation surface ends up getting ununiform. Accordingly, in the microscope unit **100** according to the present embodiment, the illumination optical system has the fly-eye lens **34** disposed therein, whereby uniformity of illumination at the observation surface is achieved.

(65) However, in the case of such a configuration being adopted, the number of optical components configuring the illumination optical system increases, so that sometimes, total length ends up increasing, and it ends up getting more difficult for downsizing to be achieved.

(66) Accordingly, in the illumination lens barrel **20** of the microscope unit **100** according to the first embodiment, downsizing of the microscope unit is achieved by a method of the kind described below.

(67) [Position Adjustment of Field Stop **36**]

(68) In the microscope unit **100** according to the first embodiment, the field stop **36** is disposed more inwardly than the outer peripheral surface of the intermediate lens barrel **22** is. Hence, it is possible for space inside the first lens barrel **21** to be utilized more effectively compared to in such a case as when, for example, the field stop **36** is disposed more outwardly than the outer peripheral surface of the intermediate lens barrel **22** (for example, more to the upper side than the lower end surface **27d** of the flange portion **27c** of the sixth sleeve portion **27** in FIG. **1**) is. This makes it possible for total length of the illumination lens barrel **20** to be configured short, whereby downsizing of the illumination lens barrel overall is achieved, and, consequently, downsizing of the microscope unit **100** is achieved.

(69) In the case of downsizing of the microscope unit **100** being achieved by such a configuration, the field stop **36** is desirably disposed brought as close as is possible to a vicinity of the reflecting mirror **37**. For example, as shown in FIG. **1**, the field stop **36** is desirably disposed more inwardly than an inner peripheral surface **28e** of the intermediate lens barrel **22**. However, if the field stop **36** ends up interfering with the light flux of the reflecting mirror **37**, then there ends up occurring deterioration in illumination performance (deterioration in uniformity of brightness or illumination) of the illumination optical system. Hence, as shown in FIG. **1**, for example, the field stop **36** is desirably sufficiently separated from the reflecting mirror **37** to prevent it from interfering with the light flux of the reflecting mirror **37**. Moreover, as mentioned above, the field stop **36** and the focal position of the objective lens **32** are configured to be in a conjugate relationship. Now, if the reflecting mirror **37** is not positioned outside a range of focal depth of the field stop **36**, then there is a risk of dirt or scratches, and so on, on a surface of the reflecting mirror **37** ending up being projected onto the imaging surface IS. Hence, the reflecting mirror **37** is desirably provided outside the range of focal depth of the field stop **36**.

(70) Note that the sixth sleeve portion **27** of the illumination lens barrel **20** according to the present embodiment has the truncated conical portion **27a** formed in its lower end portion in FIG. **1**, and a lower end portion of this truncated conical portion **27a** configures the field stop **36**. Such a field stop **36** can be produced by a single chuck by a lathe device, or the like, for example, at a time of the sixth sleeve portion **27** being formed, so is capable of being manufactured with high accuracy while manufacturing man-hours are suppressed. Moreover, it is capable of being realized without

components for a separate stop, or the like, being used.

(71) Moreover, as mentioned above, in the microscope unit **100** according to the present embodiment, the illumination lens barrel **20** is capable of being rotated with respect to the main lens barrel **10**. Now, depending on configuration of the field stop, there is a risk that when the illumination lens barrel **20** is rotated with respect to the main lens barrel **10**, a center position of the field stop will end up deviating from the optical axis of the illumination optical system. Now, the field stop **36** according to the present embodiment is configured from part of the sixth sleeve portion **27** housing the first relay lens group **35**. Hence, it is possible for a positional relationship of the center position of the field stop and the optical axis of the illumination optical system to be suitably maintained, even when the illumination lens barrel **20** has been rotated with respect to the main lens barrel **10**.

(72) [Design of First Relay Lens Group **35**]

(73) The first relay lens group **35** according to the present embodiment comprises: the first lens **35a** consisting of a plano-convex lens having its convex surface directed to the lower side in FIG. **1**; and the second lens **35b** consisting of a plano-convex lens having its convex surface directed to the upper side in FIG. **1**. Such a configuration makes it possible for a focal distance from an emitting end of light of the first relay lens group **35** to a position of the field stop **36** to be configured shorter, and for even more downsizing of the microscope unit to thereby be achieved, compared to when, for example, a single relay lens has been disposed. Moreover, due to there being adopted a plano-convex lens rather than an aspherical lens, it is possible for a reduction in manufacturing costs to be realized. Note that from a viewpoint of downsizing of the illumination lens barrel overall, it is suitable for distance in an optical path from the first relay lens group **35** to the second relay lens **38** to be set so as to be no more than five times an outer diameter of the second relay lens **38**, for example. Doing so makes it possible for total length of the illumination lens barrel to be designed shorter.

Second Embodiment

(74) FIGS. **7a** and **7b** are schematic explanatory diagrams showing configuration of a microscope unit **100** according to a second embodiment. FIG. **7a** shows overall configuration of the microscope unit **100**. In FIG. **7a**, part of the microscope unit **100** is shown as a cross-sectional view, and part of the microscope unit **100** is shown as a transparency view. FIG. **7b** shows in enlarged form configuration of part of the microscope unit **100**. Note that in the description hereafter, configuring elements the same as or corresponding to those of the first embodiment will be assigned with the same symbols as in the first embodiment, and duplicated descriptions thereof will be omitted.

(75) As shown in FIG. **7a**, the microscope unit **100** according to the second embodiment differs from the microscope unit **100** according to the first embodiment in that structure of a field stop **36A** in the illumination lens barrel **20** differs from structure of the field stop **36** formed in the truncated conical portion **27a** of the sixth sleeve portion **27** of the first lens barrel **21**.

(76) That is, in the microscope unit **100** according to the second embodiment, for example, as shown in FIG. **7b**, a mirror frame member **40A** is housed inside the intermediate lens barrel **22**. Moreover, the field stop **36A** according to the second embodiment is disposed in the mirror frame member **40A**. That is, the field stop **36A** is provided as a separate component rather than being formed as part of the sixth sleeve portion **27** in the first lens barrel **21** of the illumination lens barrel **20** according to the first embodiment.

(77) This field stop **36A**, which consists of an annular round-shaped flat plate member, is adhered via the likes of an adhesive to an upper end surface of a flange-like fitting portion **43a** that is formed on a side of a lower opening end of the first lens barrel side opening portion **43** formed in the frame body **49** of the mirror frame member **40A**, in such a manner that the fitting portion **43a** projects toward a center of the first lens barrel side opening portion **43**. As a result, the field stop **36A** is disposed more inwardly than an outer peripheral surface **49a** of the frame body **49** of the

mirror frame member **40A**, similarly to the field stop **36** in the first embodiment.

(78) Note that in the second embodiment, a sixth sleeve portion **27A** of the first lens barrel **21** does not have formed therein the truncated conical portion **27a** that had been present more to the lower side than the outer peripheral portion **27e** had been, hence it is possible for structure of the sixth sleeve portion **27A** to be more simply configured. In the microscope unit **100** according to the second embodiment too, downsizing of the illumination lens barrel **20** is realizable, similarly to in the first embodiment.

(79) [Outward Appearance of Microscope Units **100** According to First Embodiment and Second Embodiment]

(80) FIGS. **8a**, **8b**, **9a**, **9b**, **9c**, **10a**, **10b**, and **10c** are explanatory diagrams showing outward appearance of the above-mentioned microscope units **100**. FIG. **8a** is a perspective view from above of the microscope unit **100**. FIG. **8b** is a perspective view from below of the microscope unit **100**. FIG. **9a** is a front view of the microscope unit **100**. FIG. **9b** is a back view (a rear view) of the microscope unit **100**. FIG. **9c** is a plan view (a top view) of the microscope unit **100**. FIG. **10a** is a bottom surface view (a bottom view) of the microscope unit **100**. FIG. **10b** is a right-side view of the microscope unit **100**. FIG. **10c** is a left-side view of the microscope unit **100**. Note that in FIGS. **8a** to **10c**, main symbols assigned to each of the previously described configuring elements of the main lens barrel **10** and illumination lens barrel **20** are not shown, and duplicated descriptions thereof will be omitted.

(81) The microscope unit **100** is to be fitted to a microscope to provide an imaging optical system and illumination optical system for observing or imaging an object of an observation surface. As mentioned above, in the microscope unit **100**, the illumination lens barrel **20** configuring the illumination optical system of the microscope is configured smaller-sized than the main lens barrel **10** configuring the imaging optical system of the microscope, and, moreover, is integrally connected to the main lens barrel **10** in a manner enabling it to revolve with respect to the main lens barrel **10** such that its attitude freely changes from the basic attitude. The main lens barrel **10** is capable of being fitted with an imaging sensor and objective lens of the microscope. The illumination lens barrel **20**, which is capable of being fitted with a variety of light sources, is connected to the main lens barrel **10**.

(82) As shown in FIGS. **8a** and **8b**, the cylindrical first sleeve portion of the main lens barrel **10** has: a device-connecting flange portion **101** of annular shape provided on an upper side of the first sleeve portion; and a first fitting flange portion **102** of rectangular shape provided on a lower side of the first sleeve portion. The cylindrical second sleeve portion of the main lens barrel **10** has: a second fitting flange portion **103** of rectangular shape provided on an upper side of the second sleeve portion; and a third fitting flange portion **104** of rectangular shape provided on a lower side of the second sleeve portion.

(83) The first sleeve portion and second sleeve portion are connected by the first fitting flange portion **102** and second fitting flange portion **103** having their positions aligned, after which, as shown in FIGS. **8a** and **9c**, places on inner sides of their four corners are each fixed by a bolt **91**. The first casing and second casing of the case of main body portion of the main lens barrel **10** are formed in shapes of boxes with chamfered corners, of rectangular shape, having substantially the same outer shapes, as shown in FIGS. **9a**, **9b**, **10b**, and **10c**.

(84) The second sleeve portion is connected to the case of main body portion by the third fitting flange portion **104** having its position aligned with an upper portion of the first casing of the case of main body portion, after which, as shown in FIGS. **8a**, **9a**, **9b**, **10b**, and **10c**, places on inner sides of its four corners are each fixed by a bolt **92**.

(85) The cylindrical third sleeve portion of the main lens barrel **10** has a fourth fitting flange portion **105** of annular shape provided on an upper side of the third sleeve portion. The third sleeve portion is connected to the case of main body portion by having its position aligned with a lower portion of the second casing of the case of main body portion, after which, as shown in FIGS. **8b**,

9a, 9b, and 10a to 10c, a certain four places thereof are each fixed by a bolt **93**. Note that the first casing and second casing of the case of main body portion are integrally configured by being fixed by a certain number of bolts **94** from a lower side of the second casing, as shown in FIG. **10a**.

(86) As shown in FIGS. **8a** and **8b**, the first lens barrel of the illumination lens barrel **20** is formed in a cylindrical shape, and the intermediate lens barrel of the illumination lens barrel **20** is formed in a rectangular tubular shape having an inclined portion. As shown in FIGS. **8a, 8b, 9a to 9c**, and **10b**, an outer diameter of the fourth sleeve portion of the first lens barrel is formed smaller than an outer diameter of the fifth sleeve portion of the first lens barrel, and the outer diameter of the fifth sleeve portion is formed smaller than an outer diameter of the coupler case of portion **26** of the first lens barrel. Moreover, an outer diameter of the sixth sleeve portion of the first lens barrel is formed with substantially the same diameter as the outer diameter of the fifth sleeve portion.

(87) As mentioned above, the first lens barrel is fixed by the fitting bolts **19** to the intermediate lens barrel, while, as shown in FIGS. **8a, 8b, 9a, 9b, 10a**, and **10b**, the intermediate lens barrel is connected to the main lens barrel by a pair of fifth fitting flange portions **106** provided on a second casing side of the case of main body portion of the main lens barrel being fixed by a certain number of bolts **95**. Moreover, the inclined portion of the intermediate lens barrel is formed in a freely attachable/detachable manner in a rounded-cornered rectangular shape, and is fixed to the intermediate lens barrel by a pair of diagonally-aligned bolts **96**, as shown in FIGS. **8b, 10a**, and **10b**.

OTHER EMBODIMENTS

(88) Each portion of the microscope unit **100** in each of the above-mentioned embodiments is capable of having its outer shape, dimensions, and so on, appropriately changed. For example, it is also possible for the case of main body portion **13** of the main lens barrel **10** to be formed in a cylindrical shape, and for an entirety from the first sleeve portion **11** to the third sleeve portion **14** to be formed in a cylindrical shape. Moreover, it is also possible for the first sleeve portion **11**, second sleeve portion **12**, and third sleeve portion **14** to be formed in a rectangular shape, and for the entirety from the first sleeve portion **11** to the third sleeve portion **14** to be formed in a rectangular (rectangular tubular) shape. Furthermore, it is also possible for outer diameters of the first through third sleeve portions **11, 12, 14** to be made different from what was mentioned above. The above changes may be applied to the illumination lens barrel **20** too. Moreover, portions of the microscope unit **100** exemplified as configurations consisting of a plurality of members may be configured from a single member, or portions of the microscope unit **100** exemplified as configurations consisting of a single member may be configured from a plurality of members.

(89) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms: furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A microscope unit comprising: a main lens barrel of an imaging optical system, the main lens barrel extending along a first optical axis and being configured to be capable of being fitted with an imaging sensor and an objective lens; an intermediate lens barrel extending along a second optical axis orthogonal to the first optical axis, and including an end in a direction in which the second optical axis extends that is connected to an outer peripheral surface of the main lens barrel; and a first lens barrel extending along a third optical axis orthogonal to the second optical axis, including an end in a direction in which the third optical axis extends that is connected to an outer peripheral

surface of the intermediate lens barrel, and that is configured to be capable of being fitted with a light source, wherein the end of the first lens barrel in the direction in which the third optical axis extends comprises a truncated conical portion formed in a truncated conical shape tapering toward the intermediate lens barrel, the truncated conical portion is inserted more inwardly than an outer peripheral surface of the intermediate lens barrel is, and configures a field stop of light irradiated from the light source.

2. The microscope unit according to claim 1, wherein the intermediate lens barrel is inserted more inwardly than an inner peripheral surface of the intermediate lens barrel is.

3. The microscope unit according to claim 2, wherein the intermediate lens barrel and the first lens barrel are configured to be capable of revolving around the second optical axis with respect to the main lens barrel.

4. The microscope unit according to claim 3, having: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member, wherein the intermediate lens barrel is inserted more inwardly than an outer peripheral surface of the mirror frame member is.

5. The microscope unit according to claim 4, further comprising an illumination optical system, the illumination optical system comprising: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays to the beam splitter light that has been reflected by the reflecting mirror, the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.

6. The microscope unit according to claim 2, having: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member, wherein the intermediate lens barrel is inserted more inwardly than an outer peripheral surface of the mirror frame member is.

7. The microscope unit according to claim 6, further comprising an illumination optical system, the illumination optical system comprising: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays light to the beam splitter that has been reflected by the reflecting mirror, the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.

8. The microscope unit according to claim 1, wherein the intermediate lens barrel and the first lens barrel are configured to be capable of revolving around the second optical axis with respect to the main lens barrel.

9. The microscope unit according to claim 8, having: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member, wherein the intermediate lens barrel is inserted more inwardly than an outer peripheral surface of the mirror frame member is.

10. The microscope unit according to claim 9, further comprising an illumination optical system, the illumination optical system comprising: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays light to the beam splitter that has been reflected by the reflecting mirror, the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.

11. The microscope unit according to claim 1, having: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member, wherein the intermediate lens barrel is inserted more inwardly than an outer peripheral surface of the mirror frame member is.

12. The microscope unit according to claim 11, further comprising an illumination optical system, the illumination optical system comprising: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays light to the beam splitter that has been reflected by the reflecting mirror, the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.
